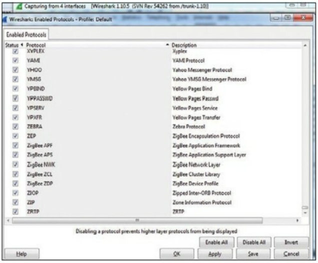

Figure 4: List of protocols with the options displayed by Wireshark

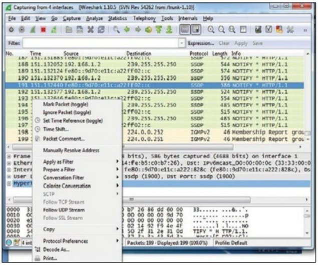

Figure 5: Analysing individual packets by right-clicking

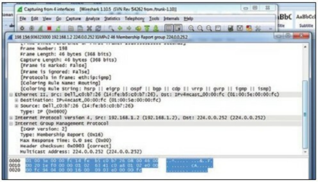

Figure 6: Individual packet information

Wireshark has also won some industry awards and recognition over the years, from the following:

- eWeek
- Infoworld
- Insecure.org system security devices survey
- Sourceforge Project of the Month in August 2010
- McAfee SiteAdvisor
- Network Protocol Analysis Award
- VoIP Monitoring Award

Wireshark is a specialised tool that automatically understands the structure and format of different networking protocols. It can intelligently parse and show the fields, along with their descriptions specified by assorted networking protocols. Wireshark makes use of *pcap* to capture the packets. This tool is able to capture packets on the types of networks that *pcap* supports.

Wireshark has a rich set of features including:

- Detailed as well as deep inspection of hundreds of protocols.
- Live capturing of packets as well as offline investigation.
- It's a cross-platform tool that can run on Windows, Linux, OS X, Solaris, FreeBSD, NetBSD, and many others, without any specific configuration.
- The captured network packets and data can be viewed via a GUI, or via the TTY-mode T Shark utility.
- It has VoIP support and analysis. VoIP calls in the captured traffic can be analysed and detected.
- Captured files compressed with gzip can be decompressed, on the fly.
- Live data can be read from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI, and others (depending on your platform).
- It has decryption support for many protocols, including IPsec, ISAKMP, Kerberos, SNMPv3, SSL/TLS, WEP, and WPA/WPA2.
- Colouring rules can be applied to the packet list for quick, intuitive analysis.
- It has the output export feature to XML, PostScript, CSV or plain text.
- Data can be captured from a live network connection or can be read from a file of already-captured packets.
- Live data can be analysed from many types of networks including Ethernet, IEEE 802.11, PPP, and loopback.
- The captured data can be edited or converted via command-line switches to the 'editcap' tool.
- The refinement in the data display can be implemented using the display filter.
- Plug-ins can be implemented and developed for new protocols.
- Raw traffic related to USB can be captured easily.

The online manual of Wireshark is available at *http://www.wireshark.org/docs/wsug_html_chunked/index.html*.

NetworkMiner

NetworkMiner is a famous network forensic analysis tool (NFAT) that can detect various system parameters including OS, hostname and open ports of network hosts through packet sniffing or by parsing a *pcap* file. The tool can extract the transmitted files from network traffic. NetworkMiner